

Unit 3. Rank, Inverse, and Determinant

1 Rank

1.1 Full-rank factorisation

Definition 3.1 (Singular and nonsingular). A square $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **nonsingular** if $\text{rank}(\mathbf{A}) = n$ and **singular** if $\text{rank}(\mathbf{A}) < n$.

Origin: H §4.4(c); A&M §4.1

Theorem 3.2 (Full-rank factorisation). If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has rank r , then

$$\mathbf{A} = \mathbf{BC}, \quad \mathbf{B} \in \mathbb{R}^{m \times r}, \quad \text{rank}(\mathbf{B}) = r, \quad \mathbf{C} \in \mathbb{R}^{r \times n}, \quad \text{rank}(\mathbf{C}) = r.$$

For $r = 0$, both factors have no columns or no rows, respectively, and their product is $\mathbf{O}$.

Origin: H Thm. 4.4.8; A&M Ex. 4.11(a)

Proof. For $r > 0$, choose basis columns $\mathbf{u}_1, \dots, \mathbf{u}_r$ of $\text{col}(\mathbf{A})$ and set $\mathbf{B} = (\mathbf{u}_1, \dots, \mathbf{u}_r)$. For each

column $\mathbf{a}_j$ of $\mathbf{A}$, unique coordinates $c_{1j}, \dots, c_{rj}$ satisfy

$$\mathbf{a}_j = \sum_{i=1}^r c_{ij} \mathbf{u}_i.$$

With $\mathbf{C} = (c_{ij})$, these n identities give $\mathbf{A} = \mathbf{BC}$. The columns of $\mathbf{B}$ are independent, so $\text{rank}(\mathbf{B}) = r$. Every row of $\mathbf{A} = \mathbf{BC}$ is a linear combination of the rows of $\mathbf{C}$; equality of row and column rank gives

$$r = \text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{C}) \leq r.$$

Thus $\text{rank}(\mathbf{C}) = r$. If $r = 0$, $\mathbf{A} = \mathbf{O}$. □

Corollary 3.3 (Rank-one expansion). Every rank- r matrix admits

$$\mathbf{A} = \sum_{k=1}^r \mathbf{u}_k \mathbf{c}'_k,$$

where every nonzero summand has rank one. No sum of fewer than r rank-one matrices can equal $\mathbf{A}$.

Origin: Course formulation from H Ex. 4.10; A&M Ex. 4.11(b); S §3.8

Proof. Write the factorisation of Theorem 3.2 as

$$\mathbf{A} = (\mathbf{u}_1, \dots, \mathbf{u}_r) \begin{pmatrix} \mathbf{c}'_1 \\ \vdots \\ \mathbf{c}'_r \end{pmatrix} = \sum_{k=1}^r \mathbf{u}_k \mathbf{c}'_k.$$

For nonzero $\mathbf{u}, \mathbf{v}$, every column of $\mathbf{u}\mathbf{v}'$ is a multiple of $\mathbf{u}$, and at least one is nonzero; hence $\text{rank}(\mathbf{u}\mathbf{v}') = 1$. If $\mathbf{A} = \sum_{k=1}^s \mathbf{A}_k$ and $\text{rank}(\mathbf{A}_k) = 1$, choose nonzero $\mathbf{u}_k \in \text{col}(\mathbf{A}_k)$. Then

$$\text{col}(\mathbf{A}_k) = \text{span}\{\mathbf{u}_k\}, \quad \text{col}(\mathbf{A}) \subseteq \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_s\},$$

so $r \leq s$. □

Example 3.1 (A 3 by 4 factorisation).

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 3 & 2 \\ 1 & 1 & 2 & 2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}}_{\mathbf{B}} \underbrace{\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 3 & 2 \end{pmatrix}}_{\mathbf{C}}.$$

The first two columns of $\mathbf{A}$ are independent, and every column belongs to their span; hence $\text{rank}(\mathbf{A}) = 2$. Also

$$\mathbf{A} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 3 & 2 \end{pmatrix}.$$

Origin: Course formulation from H Thm. 4.4.8

1.2 Products and sums

Theorem 3.4 (Rank of a product). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$,

$$\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}.$$

Moreover,

$$\begin{aligned} \text{rank}(\mathbf{AB}) &= \text{rank}(\mathbf{B}), \text{ if } \mathbf{A} \text{ has full column rank,} \\ \text{rank}(\mathbf{AB}) &= \text{rank}(\mathbf{A}), \text{ if } \mathbf{B} \text{ has full row rank.} \end{aligned}$$

Origin: H Cor. 4.4.5 and Lemma 8.3.2; A&M Ex. 4.15 and 4.25(c)–(d)

Proof. Since $\text{col}(\mathbf{AB}) \subseteq \text{col}(\mathbf{A})$, $\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A})$. Applying this result to $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ gives $\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{B})$.

If $\mathbf{A}$ has full column rank, then $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$ and

$$\mathbf{ABx} = \mathbf{0} \iff \mathbf{Bx} \in \text{null}(\mathbf{A}) \iff \mathbf{Bx} = \mathbf{0}.$$

Thus $\text{null}(\mathbf{AB}) = \text{null}(\mathbf{B})$; rank–nullity gives equal ranks. If $\mathbf{B}$ has full row rank, then $\text{col}(\mathbf{B}) = \mathbb{R}^n$,

so

$$\operatorname{col}(\mathbf{AB}) = \{\mathbf{Ay} : \mathbf{y} \in \operatorname{col}(\mathbf{B})\} = \{\mathbf{Ay} : \mathbf{y} \in \mathbb{R}^n\} = \operatorname{col}(\mathbf{A}).$$

□

Theorem 3.5 (Sylvester's rank inequality). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$,

$$\operatorname{rank}(\mathbf{AB}) \geq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}) - n.$$

More precisely,

$$\operatorname{rank}(\mathbf{AB}) = \operatorname{rank}(\mathbf{B}) - \dim(\operatorname{col}(\mathbf{B}) \cap \operatorname{null}(\mathbf{A})).$$

Origin: H Cor. 17.5.2 and Eq. 17.5.17; A&M Ex. 5.47(b)

Proof. Define $S : \operatorname{col}(\mathbf{B}) \rightarrow \mathbb{R}^m$ by $S(\mathbf{y}) = \mathbf{Ay}$. Then

$$\operatorname{im}(S) = \{\mathbf{ABx} : \mathbf{x} \in \mathbb{R}^p\} = \operatorname{col}(\mathbf{AB}), \quad \ker(S) = \operatorname{col}(\mathbf{B}) \cap \operatorname{null}(\mathbf{A}).$$

Rank-nullity for S gives the equality. Since

$$\dim(\operatorname{col}(\mathbf{B}) \cap \operatorname{null}(\mathbf{A})) \leq \dim \operatorname{null}(\mathbf{A}) = n - \operatorname{rank}(\mathbf{A}),$$

the inequality follows. □

Theorem 3.6 (Ranks of a sum and a difference). For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$,

$$\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}), \quad |\text{rank}(\mathbf{A}) - \text{rank}(\mathbf{B})| \leq \text{rank}(\mathbf{A} - \mathbf{B}).$$

Origin: H Cor. 4.5.9; A&M Ex. 4.14

Proof. Every column of $\mathbf{A} + \mathbf{B}$ lies in $\text{col}(\mathbf{A}) + \text{col}(\mathbf{B})$; hence the dimension formula gives

$$\text{rank}(\mathbf{A} + \mathbf{B}) \leq \dim(\text{col}(\mathbf{A}) + \text{col}(\mathbf{B})) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}).$$

Apply this inequality to $\mathbf{A} = (\mathbf{A} - \mathbf{B}) + \mathbf{B}$ and $\mathbf{B} = -(\mathbf{A} - \mathbf{B}) + \mathbf{A}$:

$$\text{rank}(\mathbf{A}) - \text{rank}(\mathbf{B}) \leq \text{rank}(\mathbf{A} - \mathbf{B}), \quad \text{rank}(\mathbf{B}) - \text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{A} - \mathbf{B}).$$

□

Example 3.2 (Sharp and strict rank bounds). For

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B}) = 1, \quad \text{rank}(\mathbf{AB}) = 0, \quad \text{rank}(\mathbf{A} + \mathbf{B}) = 2.$$

Thus the product upper bound can be strict, while Sylvester's lower bound and the sum upper bound are attained.

Origin: Course formulation from A&M Ex. 4.14–4.15

1.3 The Gram matrix

Theorem 3.7 (Gram rank theorem). For every $\mathbf{A} \in \mathbb{R}^{m \times n}$,

$$\text{null}(\mathbf{A}'\mathbf{A}) = \text{null}(\mathbf{A}), \quad \text{rank}(\mathbf{A}'\mathbf{A}) = \text{rank}(\mathbf{A}), \quad \text{col}(\mathbf{A}'\mathbf{A}) = \text{col}(\mathbf{A}').$$

The transposed statements give $\text{rank}(\mathbf{AA}') = \text{rank}(\mathbf{A})$ and $\text{col}(\mathbf{AA}') = \text{col}(\mathbf{A})$.

Origin: H Cor. 7.4.5; A&M Ex. 4.13

Proof. For $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{Ax} = \mathbf{0} \implies \mathbf{A}'\mathbf{Ax} = \mathbf{0}.$$

Conversely,

$$\mathbf{A}'\mathbf{Ax} = \mathbf{0} \implies 0 = \mathbf{x}'\mathbf{A}'\mathbf{Ax} = (\mathbf{Ax})'(\mathbf{Ax}) = \|\mathbf{Ax}\|^2 \implies \mathbf{Ax} = \mathbf{0}.$$

Equal nullities and rank–nullity give equal ranks. Since $\text{col}(\mathbf{A}'\mathbf{A}) \subseteq \text{col}(\mathbf{A}')$ and the two spaces have equal dimension, they are equal. Replace $\mathbf{A}$ by $\mathbf{A}'$ for the remaining identities. $\square$

Corollary 3.8 (Full column rank and the Gram matrix). For $\mathbf{A} \in \mathbb{R}^{m \times n}$,

$$\mathbf{A} \text{ has full column rank} \iff \mathbf{A}'\mathbf{A} \text{ is nonsingular.}$$

Similarly, $\mathbf{A}$ has full row rank if and only if $\mathbf{AA}'$ is nonsingular.

Origin: H Cor. 7.4.5–7.4.6; A&M Ex. 4.13(d)

Proof. The square matrix $\mathbf{A}'\mathbf{A}$ has order n . By Theorem 3.7,

$$\text{rank}(\mathbf{A}'\mathbf{A}) = n \iff \text{rank}(\mathbf{A}) = n.$$

Apply the same argument to $\mathbf{A}'$ for the row statement. □

Example 3.3 (A rank-deficient Gram matrix). Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{X}'\mathbf{X} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 6 \end{pmatrix}.$$

The third column of $\mathbf{X}$ is the sum of the first two, and

$$\mathbf{X} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}, \quad \mathbf{X}'\mathbf{X} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}.$$

The first two columns of $\mathbf{X}$ are independent, so both matrices have rank 2.

2 The inverse

2.1 Existence, uniqueness, and algebra

Definition 3.9 (One-sided and two-sided inverses). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix $\mathbf{R} \in \mathbb{R}^{n \times m}$ is a **right inverse** if $\mathbf{AR} = \mathbf{I}_m$, and a matrix $\mathbf{L} \in \mathbb{R}^{n \times m}$ is a **left inverse** if $\mathbf{LA} = \mathbf{I}_n$. If one matrix $\mathbf{B}$ is both, then $\mathbf{A}$ is **invertible** and $\mathbf{B} = \mathbf{A}^{-1}$.

Origin: H §8.1; A&M §4.2

Theorem 3.10 (Existence of one-sided inverses). For $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{A} \text{ has a right inverse} \iff \text{rank}(\mathbf{A}) = m,$$

$$\mathbf{A} \text{ has a left inverse} \iff \text{rank}(\mathbf{A}) = n.$$

Consequently, a right inverse requires $m \leq n$, and a left inverse requires $n \leq m$.

Origin: H Lemma 8.1.1

Proof. If $\mathbf{AR} = \mathbf{I}_m$, then

$$m = \text{rank}(\mathbf{I}_m) = \text{rank}(\mathbf{AR}) \leq \text{rank}(\mathbf{A}) \leq m.$$

Conversely, if $\text{rank}(\mathbf{A}) = m$, each $\mathbf{e}_j \in \mathbb{R}^m$ has the form $\mathbf{e}_j = \mathbf{A}\mathbf{r}_j$. Then $\mathbf{R} = (\mathbf{r}_1, \dots, \mathbf{r}_m)$ satisfies $\mathbf{A}\mathbf{R} = \mathbf{I}_m$. The left-inverse assertion follows by applying the right-inverse assertion to $\mathbf{A}'$ and transposing. $\square$

Theorem 3.11 (Square inverse theorem). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, the following are equivalent:

- (i) $\mathbf{A}$ is nonsingular;
- (ii) $\mathbf{A}$ has a right inverse;
- (iii) $\mathbf{A}$ has a left inverse;
- (iv) $\mathbf{A}$ is invertible.

Every one-sided inverse is the unique two-sided inverse.

Origin: H Lemma 8.1.3 and Thm. 8.1.4; A&M Ex. 4.20–4.21

Proof. Theorem 3.10 gives (i) $\iff$ (ii) $\iff$ (iii). Suppose $\mathbf{A}\mathbf{R} = \mathbf{I}_n$. Then $\mathbf{A}$ also has a left inverse $\mathbf{L}$, and

$$\mathbf{L} = \mathbf{L}(\mathbf{A}\mathbf{R}) = (\mathbf{L}\mathbf{A})\mathbf{R} = \mathbf{R}.$$

Thus $\mathbf{R}\mathbf{A} = \mathbf{I}_n$, proving (ii) $\implies$ (iv); the left-sided case is symmetric. If $\mathbf{B}$ and $\mathbf{C}$ are inverses,

then

$$\mathbf{B} = \mathbf{B}(\mathbf{A}\mathbf{C}) = (\mathbf{B}\mathbf{A})\mathbf{C} = \mathbf{C}.$$

□

Theorem 3.12 (Inverse identities). For invertible square matrices $\mathbf{A}, \mathbf{B}$ of the same order and $c \neq 0$,

$$\begin{aligned}(c\mathbf{A})^{-1} &= c^{-1}\mathbf{A}^{-1}, & (\mathbf{A}')^{-1} &= (\mathbf{A}^{-1})', & (\mathbf{A}^{-1})^{-1} &= \mathbf{A}, \\ (\mathbf{A}\mathbf{B})^{-1} &= \mathbf{B}^{-1}\mathbf{A}^{-1}.\end{aligned}$$

Origin: H Eqs. 8.2.1–8.2.8; S Prop. 3.6; A&M Ex. 4.22

Proof. Each proposed inverse is verified on both sides:

$$\begin{aligned}(c\mathbf{A})(c^{-1}\mathbf{A}^{-1}) &= \mathbf{I} = (c^{-1}\mathbf{A}^{-1})(c\mathbf{A}), \\ \mathbf{A}'(\mathbf{A}^{-1})' &= (\mathbf{A}^{-1}\mathbf{A})' = \mathbf{I}, & (\mathbf{A}^{-1})'\mathbf{A}' &= (\mathbf{A}\mathbf{A}^{-1})' = \mathbf{I}, \\ (\mathbf{A}\mathbf{B})(\mathbf{B}^{-1}\mathbf{A}^{-1}) &= \mathbf{I} = (\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{A}\mathbf{B}).\end{aligned}$$

The defining equalities for $\mathbf{A}^{-1}$ prove the third identity.

□

Example 3.4 (Exact inverses of orders 2 and 3).

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{I}_2.$$

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 2 & 1 \\ 3 & 0 & 1 \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} 2 & 1 & -1 \\ 2 & 2 & -2 \\ -6 & -3 & 5 \end{pmatrix}, \quad \mathbf{AM} = \mathbf{MA} = 2\mathbf{I}_3.$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{2}\mathbf{M}.$$

Origin: Course formulation from H Eq. 8.1.2

2.2 Orthogonal matrices

Definition 3.13 (Orthogonal matrix). A real square matrix $\mathbf{Q}$ is **orthogonal** if

$$\mathbf{Q}'\mathbf{Q} = \mathbf{I}.$$

Origin: H §8.4(a); A&M Ex. 4.27

Theorem 3.14 (Orthogonal inverse and orthonormal rows). For a square matrix $\mathbf{Q}$, the following are equivalent:

$$\mathbf{Q}'\mathbf{Q} = \mathbf{I}, \quad \mathbf{Q}\mathbf{Q}' = \mathbf{I}, \quad \mathbf{Q}^{-1} = \mathbf{Q}'.$$

Thus the columns of $\mathbf{Q}$ are orthonormal if and only if its rows are orthonormal.

Origin: H §8.4(a); A&M Ex. 4.27(a)–(c)

Proof. The equality $\mathbf{Q}'\mathbf{Q} = \mathbf{I}$ makes $\mathbf{Q}'$ a left inverse. The square inverse theorem gives $\mathbf{Q}^{-1} = \mathbf{Q}'$ and hence $\mathbf{Q}\mathbf{Q}' = \mathbf{I}$. The reverse implication is symmetric. The (i, j) entries of $\mathbf{Q}'\mathbf{Q}$ and $\mathbf{Q}\mathbf{Q}'$ are, respectively, inner products of columns and of rows. $\square$

Theorem 3.15 (Orthogonal invariance). If $\mathbf{Q}$ is orthogonal, then for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$(\mathbf{Q}\mathbf{x})'(\mathbf{Q}\mathbf{y}) = \mathbf{x}'\mathbf{y}, \quad \|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

The transpose and every finite product of orthogonal matrices are orthogonal.

Origin: H Lemma 8.4.1 and §8.4(a); A&M Ex. 4.27(c)–(d)

Proof.

$$(\mathbf{Q}\mathbf{x})'(\mathbf{Q}\mathbf{y}) = \mathbf{x}'\mathbf{Q}'\mathbf{Q}\mathbf{y} = \mathbf{x}'\mathbf{y}.$$

Set $\mathbf{y} = \mathbf{x}$ and take nonnegative square roots for the norm identity. Also

$$\mathbf{Q}\mathbf{Q}' = \mathbf{I}, \quad (\mathbf{Q}_1\mathbf{Q}_2)'(\mathbf{Q}_1\mathbf{Q}_2) = \mathbf{Q}_2'\mathbf{Q}_1'\mathbf{Q}_1\mathbf{Q}_2 = \mathbf{I};$$

induction gives the finite-product statement. □

Example 3.5 (A rational orthogonal matrix).

$$\mathbf{Q} = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad \mathbf{Q}'\mathbf{Q} = \mathbf{I}_2, \quad \mathbf{Q} \begin{pmatrix} 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

$$\left\| \begin{pmatrix} 5 \\ 0 \end{pmatrix} \right\| = \left\| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = 5.$$

Origin: Course formulation from H §8.4(a)

2.3 Equivalent nonsingularity criteria

Theorem 3.16 (Nonsingularity criteria). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, the following are equivalent:

- (i) $\text{rank}(\mathbf{A}) = n$;
- (ii) $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$;
- (iii) the columns of $\mathbf{A}$ are linearly independent;
- (iv) the columns of $\mathbf{A}$ form a basis of $\mathbb{R}^n$;
- (v) $\text{col}(\mathbf{A}) = \mathbb{R}^n$;

- (vi) $\mathbf{Ax} = \mathbf{b}$ has exactly one solution for every $\mathbf{b} \in \mathbb{R}^n$;
- (vii) $\mathbf{A}$ is invertible.

The same independence and basis criteria hold for the rows.

Origin: H §4.4(c) and Thm. 8.1.4; A&M Ex. 4.21

Proof. Rank-nullity gives (i) $\iff$ (ii). The definition of linear independence gives (ii) $\iff$ (iii). A list of n vectors in $\mathbb{R}^n$ is independent if and only if it is a basis, so (iii) $\iff$ (iv); the definition of a basis gives (iv) $\iff$ (v).

Condition (v) gives existence in (vi), while (ii) gives uniqueness because

$$\mathbf{Ax}_1 = \mathbf{Ax}_2 \iff \mathbf{x}_1 - \mathbf{x}_2 \in \text{null}(\mathbf{A}).$$

Condition (vi) implies (v), and (i) $\iff$ (v) by dimension. Finally, (i) $\iff$ (vii) is the square inverse theorem. Apply (iii) to $\mathbf{A}'$ and use $\text{rank}(\mathbf{A}') = \text{rank}(\mathbf{A})$ for the row criteria. $\square$

Corollary 3.17 (Cancellation by full-rank factors). For conformable matrices:

$$\mathbf{CA} = \mathbf{CB}, \mathbf{C} \text{ full column rank} \implies \mathbf{A} = \mathbf{B},$$

$$\mathbf{AD} = \mathbf{BD}, \mathbf{D} \text{ full row rank} \implies \mathbf{A} = \mathbf{B}.$$

Origin: H Lemma 8.3.1

Proof. A full-column-rank $\mathbf{C}$ has a left inverse $\mathbf{L}$, and a full-row-rank $\mathbf{D}$ has a right inverse $\mathbf{R}$. Hence

$$\mathbf{CA} = \mathbf{CB} \implies \mathbf{A} = \mathbf{LCA} = \mathbf{LCB} = \mathbf{B},$$

$$\mathbf{AD} = \mathbf{BD} \implies \mathbf{A} = \mathbf{ADR} = \mathbf{BDR} = \mathbf{B}.$$



3 The determinant

3.1 Definition and elementary properties

Definition 3.18 (Permutation, sign, and determinant). A **permutation** of $\{1, \dots, n\}$ is a bijection from this set to itself; let S_n be the set of all such maps. For $\sigma \in S_n$, set

$$N(\sigma) = \#\{(i, j) : i < j, \sigma(i) > \sigma(j)\}, \quad \text{sgn}(\sigma) = (-1)^{N(\sigma)}.$$

For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{n \times n}$, define

$$\det(\mathbf{A}) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}.$$

Also set $\det(\mathbf{I}_0) = 1$ for the empty matrix.

Origin: H §13.1(a)–(b); A&M §4.3; S §3.1

Example 3.6 (Determinants of orders 2 and 3).

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

For the matrix $\mathbf{A}$ in Example 3.4,

$$\det(\mathbf{A}) = 2(2)(1) + (-1)(1)(3) - (-1)(1)(1) = 2.$$

Origin: H Eqs. 13.1.4–13.1.5; A&M Ex. 4.29; S §3.1

Theorem 3.19 (Multilinearity and alternation). As a function of the columns, $\det(\mathbf{A})$ is linear in each column while the others are fixed. Interchanging two columns multiplies the determinant by -1 ; a matrix with two equal columns has determinant 0. The same statements hold for rows.

Origin: H §13.1(b), Lemma 13.2.1, and Thm. 13.2.6; S Prop. 3.2; A&M Ex. 4.30, 4.32–4.33, and 4.39

Proof. In every term of Definition 3.18, exactly one factor comes from each column, proving linearity. Swapping two entries in the list $\sigma(1), \dots, \sigma(n)$ toggles their mutual inversion; for every entry between their positions, the two associated inversion indicators change by 0 or 2. Therefore the inversion parity changes. If columns r, s are interchanged, pairing σ with $(r\ s) \circ \sigma$ shows that the determinant changes sign. If the two columns are equal, the interchange leaves the matrix fixed, so $\det(\mathbf{A}) = -\det(\mathbf{A})$ and $\det(\mathbf{A}) = 0$.

For the transpose, the term indexed by σ becomes the term indexed by σ^{-1} . An inversion $i < j$, $\sigma(i) > \sigma(j)$ maps to the inversion $\sigma(j) < \sigma(i)$, $\sigma^{-1}(\sigma(j)) > \sigma^{-1}(\sigma(i))$. This is a bijection, so $\text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma)$ and $\det(\mathbf{A}') = \det(\mathbf{A})$. Apply the column statements to $\mathbf{A}'$ to obtain the row statements. $\square$

Corollary 3.20 (Elementary operations). For a square matrix:

- (i) scaling one row or column by c scales the determinant by c ;
- (ii) adding a multiple of one row or column to another leaves the determinant fixed;
- (iii) interchanging two rows or columns changes its sign;
- (iv) $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$ and $\det(\mathbf{A}') = \det(\mathbf{A})$.

Origin: H Lemmas 13.2.1–13.2.2 and Thm. 13.2.10; A&M Ex. 4.30, 4.32, 4.34–4.35

Proof. Parts (i), (iii), and the transpose identity are in Theorem 3.19. For columns $\mathbf{a}_i, \mathbf{a}_j$ with $i \neq j$, multilinearity and alternation give

$$\det(\dots, \mathbf{a}_i + c\mathbf{a}_j, \dots, \mathbf{a}_j, \dots) = \det(\mathbf{A}) + c \det(\dots, \mathbf{a}_j, \dots, \mathbf{a}_j, \dots) = \det(\mathbf{A}).$$

Apply (i) to all n rows for $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$. □

Theorem 3.21 (Triangular determinant). If $\mathbf{T} = (t_{ij}) \in \mathbb{R}^{n \times n}$ is upper or lower triangular, then

$$\det(\mathbf{T}) = \prod_{i=1}^n t_{ii}.$$

Origin: H Lemma 13.1.1; S Prop. 3.1; A&M Ex. 4.38

Proof. For upper triangular $\mathbf{T}$, a nonzero term in the permutation sum requires $\sigma(i) \geq i$ for every i . Since $\sum_i \sigma(i) = \sum_i i$, every inequality is equality. Only $\sigma(i) = i$ remains, with sign 1, giving $\prod_i t_{ii}$. Transpose for the lower triangular case. □

3.2 Cofactors, products, and the inverse

Definition 3.22 (Minor, cofactor, and adjugate). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, let $\mathbf{A}_{ij}$ be obtained by deleting row i and column j . Define

$$M_{ij} = \det(\mathbf{A}_{ij}), \quad C_{ij} = (-1)^{i+j} M_{ij}, \quad \text{adj}(\mathbf{A}) = (C_{ij})'.$$

M_{ij} and C_{ij} are the **minor** and **cofactor** of a_{ij} ; $\text{adj}(\mathbf{A})$ is the **adjugate**.

Origin: H §13.5; S §3.2.1; A&M §4.3

Theorem 3.23 (Laplace expansion and adjugate identity). For each row i and column j ,

$$\det(\mathbf{A}) = \sum_{k=1}^n a_{ik} C_{ik} = \sum_{k=1}^n a_{kj} C_{kj}.$$

Moreover,

$$\mathbf{A} \text{adj}(\mathbf{A}) = \text{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}_n.$$

Origin: H Thms. 13.5.1–13.5.3; S Prop. 3.3; A&M Ex. 4.36–4.37 and 4.46(a)

Proof. Fix row i . Group the permutation sum by $k = \sigma(i)$. Deleting the chosen row and column leaves the permutation sum for $\det(\mathbf{A}_{ik})$. Moving row i and column k to the first positions uses $(i - 1) + (k - 1)$ interchanges and contributes $(-1)^{i+k}$. Hence $\det(\mathbf{A}) = \sum_k a_{ik} C_{ik}$. Apply this row formula to $\mathbf{A}'$ for the column formula.

The (i, j) entry of $\mathbf{A} \operatorname{adj}(\mathbf{A})$ is $\sum_k a_{ik} C_{jk}$. For $i = j$ this is the row expansion. For $i \neq j$ it is the row- j expansion of a matrix whose row j has been replaced by row i ; that matrix has two equal rows, so the sum is 0. Thus $\mathbf{A} \operatorname{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}_n$. The column argument gives the reversed product. $\square$

Corollary 3.24 (Adjugate inverse and Cramer's rule). If $\det(\mathbf{A}) \neq 0$, then

$$\mathbf{A}^{-1} = \frac{\operatorname{adj}(\mathbf{A})}{\det(\mathbf{A})}.$$

For $\mathbf{Ax} = \mathbf{b}$, let $\mathbf{A}_j(\mathbf{b})$ replace column j of $\mathbf{A}$ by $\mathbf{b}$. Then

$$x_j = \frac{\det(\mathbf{A}_j(\mathbf{b}))}{\det(\mathbf{A})}, \quad j = 1, \dots, n.$$

Origin: H Cor. 13.5.4 and Ex. 13.11; S Prop. 3.10; A&M Ex. 4.46(b)

Proof. Divide the adjugate identity by $\det(\mathbf{A})$. Then

$$x_j = \frac{1}{\det(\mathbf{A})} \sum_{i=1}^n C_{ij} b_i.$$

The numerator is the Laplace expansion of $\det(\mathbf{A}_j(\mathbf{b}))$ along column j . □

Theorem 3.25 (Multiplicativity). For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$,

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

Origin: H Thm. 13.3.4; S Thm. 3.1; A&M Ex. 4.42

Proof. Write $\mathbf{A} = (\mathbf{a}_1, \dots, \mathbf{a}_n)$ and let b_{ij} be the entries of $\mathbf{B}$. Column j of $\mathbf{AB}$ is $\sum_i b_{ij} \mathbf{a}_i$. Multilinearity gives

$$\det(\mathbf{AB}) = \sum_{i_1, \dots, i_n} b_{i_1 1} \cdots b_{i_n n} \det(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_n}).$$

Repeated indices contribute 0. For $(i_1, \dots, i_n) = (\sigma(1), \dots, \sigma(n))$, alternation gives

$$\det(\mathbf{a}_{\sigma(1)}, \dots, \mathbf{a}_{\sigma(n)}) = \operatorname{sgn}(\sigma) \det(\mathbf{A}).$$

Factoring $\det(\mathbf{A})$ leaves the permutation formula for $\det(\mathbf{B})$ after the substitution $\tau = \sigma^{-1}$, since $\operatorname{sgn}(\tau) = \operatorname{sgn}(\sigma)$. $\square$

Theorem 3.26 (Determinant criterion). For $\mathbf{A} \in \mathbb{R}^{n \times n}$,

$$\det(\mathbf{A}) \neq 0 \iff \mathbf{A} \text{ is nonsingular} \iff \mathbf{A} \text{ is invertible.}$$

If these hold, $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$.

Origin: H Thm. 13.3.7; S Prop. 3.5; A&M Ex. 4.43–4.44

Proof. If $\det(\mathbf{A}) \neq 0$, Corollary 3.24 constructs the inverse. If $\mathbf{A}$ is invertible, multiplicativity gives

$$1 = \det(\mathbf{I}_n) = \det(\mathbf{A}\mathbf{A}^{-1}) = \det(\mathbf{A})\det(\mathbf{A}^{-1}),$$

which proves both the nonzero criterion and the reciprocal identity. The square inverse theorem identifies invertibility with nonsingularity. $\square$

3.3 Minors and determinant structure

Theorem 3.27 (Rank from minors). If $\mathbf{A} \neq \mathbf{O}$, then $\text{rank}(\mathbf{A})$ is the largest k for which $\mathbf{A}$ has a k by k submatrix with nonzero determinant. The null matrix has rank 0 and all its positive-order minors vanish.

Origin: H Thm. 4.4.10

Proof. Let $\mathbf{M}$ retain rows $i_1, \dots, i_k$ and columns $j_1, \dots, j_k$ of $\mathbf{A}$. Define $\mathbf{E} \in \mathbb{R}^{k \times m}$ and $\mathbf{F} \in \mathbb{R}^{n \times k}$ by

$$(\mathbf{E})_{ai} = \begin{cases} 1, & i = i_a, \\ 0, & i \neq i_a, \end{cases} \quad (\mathbf{F})_{jb} = \begin{cases} 1, & j = j_b, \\ 0, & j \neq j_b. \end{cases}$$

Then $\mathbf{M} = \mathbf{EAF}$, so

$$\det(\mathbf{M}) \neq 0 \implies k = \text{rank}(\mathbf{M}) \leq \text{rank}(\mathbf{A}).$$

Let $r = \text{rank}(\mathbf{A}) > 0$. Choose r basis columns and put them in $\mathbf{B} \in \mathbb{R}^{m \times r}$. Let $\mathbf{E}$ select r independent rows of $\mathbf{A}$. In the factorisation $\mathbf{A} = \mathbf{BC}$,

$$\mathbf{z}'\mathbf{EB} = \mathbf{0}' \implies \mathbf{z}'\mathbf{EA} = \mathbf{z}'\mathbf{EBC} = \mathbf{0}' \implies \mathbf{z} = \mathbf{0}.$$

Thus their r by r intersection $\mathbf{M} = \mathbf{EB}$ has rank r , so $\det(\mathbf{M}) \neq 0$ by Theorem 3.26. $\square$

Corollary 3.28 (Determinants under transpose, Gram formation, and rotation). For square $\mathbf{A}$ and orthogonal $\mathbf{Q}$,

$$\det(\mathbf{A}') = \det(\mathbf{A}), \quad \det(\mathbf{A}'\mathbf{A}) = \det(\mathbf{A})^2, \quad \det(\mathbf{Q}) \in \{-1, 1\}.$$

Origin: H Lemma 13.2.1 and Cor. 13.3.5–13.3.6; S §3.3

Proof. The transpose identity was proved in Theorem 3.19. Multiplicativity then gives $\det(\mathbf{A}'\mathbf{A}) = \det(\mathbf{A})^2$. For orthogonal $\mathbf{Q}$,

$$1 = \det(\mathbf{I}) = \det(\mathbf{Q}'\mathbf{Q}) = \det(\mathbf{Q})^2.$$



Theorem 3.29 (Vandermonde determinant). For

$$\mathbf{V}(x_1, \dots, x_n) = \begin{pmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \cdots & x_n^{n-1} \end{pmatrix},$$

$$\det \mathbf{V}(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

Hence $\mathbf{V}$ is nonsingular if and only if $x_1, \dots, x_n$ are distinct.

Origin: H §13.6; A&M Ex. 4.41; S Ex. 3.16–3.17

Proof. Induct on n ; the result is 1 for $n = 1$. Let $\mathbf{T}$ have diagonal entries 1, superdiagonal entries $-x_n$, and all other entries 0. Then $\det(\mathbf{T}) = 1$ and

$$\mathbf{VT} = \begin{pmatrix} 1 & x_1 - x_n & x_1(x_1 - x_n) & \cdots & x_1^{n-2}(x_1 - x_n) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n-1} - x_n & x_{n-1}(x_{n-1} - x_n) & \cdots & x_{n-1}^{n-2}(x_{n-1} - x_n) \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Let $\mathbf{D} = \text{diag}(x_1 - x_n, \dots, x_{n-1} - x_n)$ and $\mathbf{W} = \mathbf{V}(x_1, \dots, x_{n-1})$. Expansion along the last row and multiplicativity give

$$\det(\mathbf{V}) = \det(\mathbf{VT}) = (-1)^{n+1} \det(\mathbf{DW}) = \prod_{i=1}^{n-1} (x_n - x_i) \det(\mathbf{W}).$$

Apply the induction hypothesis to $\det(\mathbf{W})$. The determinant criterion gives the final equivalence. $\square$

Corollary 3.30 (Unique polynomial interpolation). If $x_1, \dots, x_n$ are distinct, then for every $y_1, \dots, y_n$ there is a unique polynomial p of degree at most $n - 1$ such that $p(x_i) = y_i$ for every i .

Origin: Course formulation from H §13.6; S Ex. 3.17

Proof. For $p(t) = \sum_{j=0}^{n-1} c_j t^j$, the interpolation equations are

$$\mathbf{V}(x_1, \dots, x_n) \mathbf{c} = \mathbf{y}.$$

Theorem 3.29 makes the coefficient matrix nonsingular; the nonsingularity criteria give one and only one $\mathbf{c}$. □

4 Computation: Numerical Checks in R

4.1 Full-rank and Gram factorisations (§1.1, §1.3)

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 3 & 2 \\ 1 & 1 & 2 & 2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}}_{\mathbf{B}} \underbrace{\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 3 & 2 \end{pmatrix}}_{\mathbf{C}}.$$

The top two rows of $\mathbf{B}$ and the left two columns of $\mathbf{C}$ equal $\mathbf{I}_2$, so both factors have rank 2. For $\mathbf{h} = (1, -3, 1, 0)'$,

$$\mathbf{A}\mathbf{h} = \mathbf{0}, \quad \mathbf{A}'\mathbf{A}\mathbf{h} = \mathbf{0}, \quad \mathbf{A} = \sum_{k=1}^2 \mathbf{b}_k \mathbf{c}'_k,$$

where $\mathbf{b}_k$ is column k of $\mathbf{B}$ and $\mathbf{c}'_k$ is row k of $\mathbf{C}$.

```

1 A <- rbind(c(1, 0, -1, 0),
2           c(0, 1, 3, 2),
3           c(1, 1, 2, 2))
4 B <- A[, 1:2, drop = FALSE]
5 C <- rbind(c(1, 0, -1, 0),
6           c(0, 1, 3, 2))
7 h <- c(1, -3, 1, 0)
8 rank_one_sum <- B[, 1] %o% C[1, ] + B[, 2] %o% C[2, ]
9 c(A_minus_BC = max(abs(A - B %*% C)),
10  A_minus_rank_one_sum = max(abs(A - rank_one_sum)),
11  Ah = max(abs(A %*% h)),
12  AtAh = max(abs(crossprod(A) %*% h)))

```

```
##          A_minus_BC A_minus_rank_one_sum
##          0          0
##          Ah          AtAh
##          0          0
```

4.2 A 3 by 3 determinant and adjugate (§3.2)

```
1 A <- rbind(c(2, -1, 0), c(1, 2, 1), c(3, 0, 1))
2 a <- A[1, 1]; b <- A[1, 2]; cc <- A[1, 3]
3 d <- A[2, 1]; e <- A[2, 2]; f <- A[2, 3]
4 g <- A[3, 1]; h <- A[3, 2]; i <- A[3, 3]
5 det_manual <- a*e*i + b*f*g + cc*d*h - cc*e*g - b*d*i - a*f*h
6 c(manual = det_manual, package = det(A), difference = det_manual - det(A))
```

```
##      manual      package difference
## 2.000e+00 2.000e+00 6.661e-16
```

```
1 adj_manual <- matrix(c(
2   e*i - f*h, cc*h - b*i, b*f - cc*e,
3   f*g - d*i, a*i - cc*g, cc*d - a*f,
4   d*h - e*g, b*g - a*h, a*e - b*d), 3, 3, byrow = TRUE)
5 c(reconstruction = max(abs(A %*% adj_manual - det_manual * diag(3))),
```

```
6 inverse = max(abs(adj_manual / det_manual - solve(A))))
```

```
## reconstruction      inverse
##      0.000e+00      8.882e-16
```

4.3 Multiplicativity and logarithmic determinants (§3.2)

```
1 set.seed(304)
2 A <- matrix(rnorm(16), 4, 4)
3 B <- matrix(rnorm(16), 4, 4)
4 c(det_product = det(A %*% B),
5   product_det = det(A) * det(B),
6   difference = det(A %*% B) - det(A) * det(B))
```

```
## det_product product_det difference
## -1.110e+00 -1.110e+00 -2.554e-14
```

```
1 D <- diag(rep(0.01, 200))
2 ld <- determinant(D, logarithm = TRUE)
3 c(det = det(D), sign = unname(ld$sign),
4   log_abs_det = unname(ld$modulus), exact_log_abs_det = 200 * log(0.01))
```

```
##                det                sign      log_abs_det
##                0                  1          -921
## exact_log_abs_det
##                -921
```

4.4 Solving against forming an inverse (§2.1)

```
1 set.seed(305)
2 n <- 180
3 A <- crossprod(matrix(rnorm(n * n), n, n)) + diag(n)
4 b <- rnorm(n)
5 t_direct <- system.time(for (k in 1:10) x_direct <- solve(A, b))["elapsed"]
6 t_inverse <- system.time(for (k in 1:10) x_inverse <- solve(A) %*% b)["elapsed"]
7 c(residual_direct = sqrt(sum((A %*% x_direct - b)^2)),
8   residual_inverse = sqrt(sum((A %*% x_inverse - b)^2)),
9   solution_difference = sqrt(sum((x_direct - x_inverse)^2)),
10  seconds_direct = unname(t_direct), seconds_inverse = unname(t_inverse))
```

```
##      residual_direct      residual_inverse solution_difference
##      1.548e-13          2.781e-13          1.029e-14
##      seconds_direct      seconds_inverse
##      6.000e-03           7.380e-01
```

5 Pitfalls

5.1 Using a determinant as a numerical rank test

The invertible matrix $0.1\mathbf{I}_{100}$ has determinant 10^{-100} . Rescaling a matrix changes its determinant by the hundredth power without changing its rank. A small determinant does not imply deficient rank.

5.2 Treating a computed zero determinant as an algebraic zero

For $\mathbf{D} = 0.01\mathbf{I}_{200}$,

$$\det(\mathbf{D}) = 10^{-400} > 0,$$

but this number is below the range of ordinary double precision. The logarithmic form $\log |\det(\mathbf{D})| = 200 \log(0.01)$ remains representable.

5.3 Reading a positive Gram diagonal as full rank

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{X}'\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Every diagonal entry of $\mathbf{X}'\mathbf{X}$ is positive, but

$$\mathbf{x} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \mathbf{0}, \quad \text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}'\mathbf{X}) = 1.$$

5.4 Computing $\mathbf{A}^{-1}\mathbf{b}$ as two operations

The identity $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ characterises the solution. It does not prescribe an algorithm. A linear-system routine computes the solution without forming every column of $\mathbf{A}^{-1}$.

6 Exercises

Exercise 3.1 (Rank factorisation and rank-one expansion). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 1 & 3 \end{pmatrix}.$$

Find a full-rank factorisation $\mathbf{A} = \mathbf{BC}$, express $\mathbf{A}$ as a sum of rank-one matrices, and prove that one rank-one summand cannot suffice.

Origin: H Thm. 4.4.8 and Ex. 4.10; A&M Ex. 4.11

Solution. The first two columns are independent, and columns 3, 4 equal $2\mathbf{a}_1 - \mathbf{a}_2$ and $\mathbf{a}_1 + 2\mathbf{a}_2$. Hence

$$\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \end{pmatrix}, \quad \mathbf{A} = \mathbf{BC}.$$

Both factors contain $\mathbf{I}_2$ as a submatrix, so both have rank 2. Thus

$$\mathbf{A} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} (1, 0, 2, 1) + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} (0, 1, -1, 2).$$

A single rank-one summand would have rank at most 1, whereas $\text{rank}(\mathbf{A}) = 2$. ■

Exercise 3.2 (Rank-one perturbation). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{u} \in \mathbb{R}^m$, and $\mathbf{v} \in \mathbb{R}^n$, prove

$$|\text{rank}(\mathbf{A} + \mathbf{u}\mathbf{v}') - \text{rank}(\mathbf{A})| \leq 1.$$

Give examples attaining -1 , 0 , and 1 .

Origin: Course formulation from S §3.8; H Cor. 4.5.9

Solution. Since $\text{rank}(\mathbf{u}\mathbf{v}') \leq 1$, Theorem 3.6 gives

$$\text{rank}(\mathbf{A} + \mathbf{u}\mathbf{v}') \leq \text{rank}(\mathbf{A}) + 1.$$

Writing $\mathbf{A} = (\mathbf{A} + \mathbf{u}\mathbf{v}') - \mathbf{u}\mathbf{v}'$ gives the reverse bound $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{A} + \mathbf{u}\mathbf{v}') + 1$. In order 2, take

$\mathbf{A}$	$\mathbf{u}\mathbf{v}'$	rank change
$\mathbf{I}_2$	$-\mathbf{e}_1\mathbf{e}_1'$	-1
$\mathbf{I}_2$	$\mathbf{e}_1\mathbf{e}_2'$	0
$\text{diag}(1, 0)$	$\mathbf{e}_2\mathbf{e}_2'$	1 .



Exercise 3.3 (A stacked Gram matrix). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times n}$, prove

$$\text{null}(\mathbf{A}'\mathbf{A} + \mathbf{B}'\mathbf{B}) = \text{null}(\mathbf{A}) \cap \text{null}(\mathbf{B}) = \text{null} \begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix},$$

and hence

$$\text{rank}(\mathbf{A}'\mathbf{A} + \mathbf{B}'\mathbf{B}) = \text{rank} \begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix}.$$

Origin: Course formulation from H Cor. 7.4.5

Solution. For $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{x}'(\mathbf{A}'\mathbf{A} + \mathbf{B}'\mathbf{B})\mathbf{x} = \|\mathbf{Ax}\|^2 + \|\mathbf{Bx}\|^2.$$

If $(\mathbf{A}'\mathbf{A} + \mathbf{B}'\mathbf{B})\mathbf{x} = \mathbf{0}$, the displayed sum is zero, so $\mathbf{Ax} = \mathbf{0}$ and $\mathbf{Bx} = \mathbf{0}$. Conversely,

$$\mathbf{Ax} = \mathbf{0}, \mathbf{Bx} = \mathbf{0} \implies (\mathbf{A}'\mathbf{A} + \mathbf{B}'\mathbf{B})\mathbf{x} = \mathbf{0}.$$

This is also equivalent to

$$\begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix} \mathbf{x} = \mathbf{0}.$$

The three matrices have n columns, so rank–nullity gives the rank identity. ■

Exercise 3.4 (Exact rank loss in a product). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$, prove

$$\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}) \iff \text{col}(\mathbf{B}) \cap \text{null}(\mathbf{A}) = \{\mathbf{0}\}.$$

Deduce that full column rank of $\mathbf{A}$ is sufficient but not necessary for equality.

Origin: H Eq. 17.5.17

Solution. The exact formula in Theorem 3.5 gives

$$\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}) - \dim(\text{col}(\mathbf{B}) \cap \text{null}(\mathbf{A})),$$

which proves the equivalence. Full column rank gives $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$. It is not necessary: with

$$\mathbf{A} = \text{diag}(1, 0), \quad \mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$\mathbf{A}$ is not full column rank, but $\mathbf{AB} = \mathbf{B}$ and both ranks are 1. ■

Exercise 3.5 (Families of right inverses). For $m, q \geq 1$, let $\mathbf{A} = (\mathbf{I}_m \ \mathbf{C}) \in \mathbb{R}^{m \times (m+q)}$. Determine all right inverses of the form

$$\mathbf{R} = \begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix}, \quad \mathbf{U} \in \mathbb{R}^{m \times m}, \quad \mathbf{V} \in \mathbb{R}^{q \times m}.$$

When $q > 0$, prove that $\mathbf{A}$ has infinitely many right inverses.

Origin: Course formulation from H Lemma 8.1.1

Solution. Block multiplication gives

$$\mathbf{AR} = \mathbf{U} + \mathbf{CV}.$$

Thus $\mathbf{AR} = \mathbf{I}_m$ exactly when

$$\mathbf{U} = \mathbf{I}_m - \mathbf{CV}, \quad \mathbf{R} = \begin{pmatrix} \mathbf{I}_m - \mathbf{CV} \\ \mathbf{V} \end{pmatrix}, \quad \mathbf{V} \in \mathbb{R}^{q \times m}.$$

If $q > 0$, distinct choices $\mathbf{V} = t\mathbf{E}_{11}$, $t \in \mathbb{R}$, give distinct right inverses. ■

Exercise 3.6 (Orthogonal recoding of data). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ and let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be orthogonal. Prove

$$(\mathbf{QX})'(\mathbf{QX}) = \mathbf{X}'\mathbf{X}, \quad \text{rank}(\mathbf{QX}) = \text{rank}(\mathbf{X}), \quad \|\mathbf{Qx}_i - \mathbf{Qx}_j\| = \|\mathbf{x}_i - \mathbf{x}_j\|$$

for any columns $\mathbf{x}_i, \mathbf{x}_j$ of $\mathbf{X}$.

Origin: Course formulation from H §8.4(a) and Lemma 8.3.2; A&M Ex. 4.13 and 4.27

Solution.

$$(\mathbf{QX})'(\mathbf{QX}) = \mathbf{X}'\mathbf{Q}'\mathbf{QX} = \mathbf{X}'\mathbf{X}.$$

Since $\mathbf{Q}$ is nonsingular, the product-rank equality gives $\text{rank}(\mathbf{QX}) = \text{rank}(\mathbf{X})$. Finally,

$$\|\mathbf{Qx}_i - \mathbf{Qx}_j\| = \|\mathbf{Q}(\mathbf{x}_i - \mathbf{x}_j)\| = \|\mathbf{x}_i - \mathbf{x}_j\|.$$



Exercise 3.7 (Commutation and inversion). If $\mathbf{AB} = \mathbf{BA}$ and $\mathbf{B}$ is invertible, prove

$$\mathbf{AB}^{-1} = \mathbf{B}^{-1}\mathbf{A}.$$

If both matrices are invertible, prove that $\mathbf{A}^{-1}$ and $\mathbf{B}^{-1}$ commute.

Origin: H Eq. 8.2.8; A&M Ex. 4.22(e)

Solution. Premultiply $\mathbf{AB} = \mathbf{BA}$ by $\mathbf{B}^{-1}$ and postmultiply by $\mathbf{B}^{-1}$:

$$\mathbf{B}^{-1}\mathbf{A} = \mathbf{AB}^{-1}.$$

If $\mathbf{A}$ is also invertible, then

$$\mathbf{A}^{-1}\mathbf{B}^{-1} = (\mathbf{BA})^{-1} = (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$$

Exercise 3.8 (Matrix determinant lemma). For $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, prove

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}') = \det(\mathbf{A}) + \mathbf{v}' \operatorname{adj}(\mathbf{A})\mathbf{u}.$$

If $\mathbf{A}$ is invertible, deduce

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}') = \det(\mathbf{A})(1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}).$$

Origin: S Prop. 3.21; H Thm. 18.1.1

Solution. Column j of $\mathbf{A} + \mathbf{u}\mathbf{v}'$ is $\mathbf{a}_j + v_j\mathbf{u}$. Multilinearity produces the term $\det(\mathbf{A})$, the n terms with one replaced column, and terms with at least two columns proportional to $\mathbf{u}$. The latter

vanish. Expanding a one-replacement determinant along its replaced column gives

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}') = \det(\mathbf{A}) + \sum_{j=1}^n v_j \sum_{i=1}^n u_i C_{ij} = \det(\mathbf{A}) + \mathbf{v}' \operatorname{adj}(\mathbf{A})\mathbf{u}.$$

For invertible $\mathbf{A}$, substitute $\operatorname{adj}(\mathbf{A}) = \det(\mathbf{A})\mathbf{A}^{-1}$. ■

Exercise 3.9 (Integer inverses). Let $\mathbf{A}$ be a square integer matrix. Prove that $\mathbf{A}^{-1}$ exists and has integer entries if and only if $\det(\mathbf{A}) = \pm 1$.

Origin: Course formulation from H Ex. 13.3; S Prop. 3.5

Solution. If $\mathbf{A}^{-1}$ is integral, then both determinants are integers and

$$\det(\mathbf{A}) \det(\mathbf{A}^{-1}) = 1,$$

so $\det(\mathbf{A}) = \pm 1$. Conversely, every cofactor of an integer matrix is an integer. If $\det(\mathbf{A}) = \pm 1$, the adjugate formula gives $\mathbf{A}^{-1} = \pm \operatorname{adj}(\mathbf{A})$, an integer matrix. ■

Exercise 3.10 (Rank of the adjugate). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, prove

$$\text{rank}(\text{adj}(\mathbf{A})) = \begin{cases} n, & \text{rank}(\mathbf{A}) = n, \\ 1, & \text{rank}(\mathbf{A}) = n - 1, \\ 0, & \text{rank}(\mathbf{A}) \leq n - 2. \end{cases}$$

Origin: Course formulation from H Ex. 13.10; A&M Ex. 4.46

Solution. If $\text{rank}(\mathbf{A}) = n$, then $\text{adj}(\mathbf{A}) = \det(\mathbf{A})\mathbf{A}^{-1}$ has rank n . If $\text{rank}(\mathbf{A}) \leq n - 2$, every $(n - 1)$ -order minor vanishes by Theorem 3.27; hence every cofactor is zero.

If $\text{rank}(\mathbf{A}) = n - 1$, some $(n - 1)$ -order minor is nonzero, so $\text{adj}(\mathbf{A}) \neq \mathbf{O}$. Since $\det(\mathbf{A}) = 0$, the adjugate identity gives $\mathbf{A} \text{adj}(\mathbf{A}) = \mathbf{O}$. Every column of $\text{adj}(\mathbf{A})$ lies in the one-dimensional space $\text{null}(\mathbf{A})$; therefore its rank is at most 1 and, being nonzero, equals 1. ■

Exercise 3.11 (Tridiagonal recurrence). Let

$$\mathbf{T}_n = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$$

Prove $\det(\mathbf{T}_n) = n + 1$.

Origin: A&M Ex. 4.40

Solution. Let $d_n = \det(\mathbf{T}_n)$ and set $d_0 = 1$. Expansion along the first row, followed by expansion of the second minor along its first column, gives

$$d_n = 2d_{n-1} - d_{n-2}, \quad d_0 = 1, \quad d_1 = 2.$$

$$d_n - d_{n-1} = d_{n-1} - d_{n-2} = \cdots = d_1 - d_0 = 1 \quad \implies \quad d_n = d_0 + n = n + 1. \quad \blacksquare$$

Exercise 3.12 (Interpolation as a linear system). Let $x_1, \dots, x_n$ be distinct. Prove that the evaluation map

$$E : \mathcal{P}_{n-1} \rightarrow \mathbb{R}^n, \quad E(p) = (p(x_1), \dots, p(x_n))',$$

is linear, one-to-one, and onto. If two polynomials of degree at most $n - 1$ agree at all x_i , prove that they are equal.

Origin: Course formulation from H §13.6; A&M Ex. 4.41; S Ex. 3.17

Solution. For $p(t) = \sum_{j=0}^{n-1} c_j t^j$ and $\mathbf{c} = (c_0, \dots, c_{n-1})'$,

$$E(p) = \mathbf{V}(x_1, \dots, x_n)\mathbf{c}.$$

The determinant of $\mathbf{V}$ is

$$\prod_{i < j} (x_j - x_i) \neq 0,$$

and E is linear by componentwise evaluation.

For each $\mathbf{y} \in \mathbb{R}^n$, the system $\mathbf{V}\mathbf{c} = \mathbf{y}$ has exactly one solution, so E is one-to-one and onto. If $p(x_i) = q(x_i)$ for all i , then $E(p - q) = \mathbf{0}$; one-to-one-ness gives $p - q = 0$. ■

Exercise 3.13 (Cauchy–Binet). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times m}$ with $m \leq n$. For $J \subseteq \{1, \dots, n\}$ with $|J| = m$, let $\mathbf{A}_{:,J}$ retain the columns in J and let $\mathbf{B}_{J,:}$ retain the rows in J , both in increasing order. Prove

$$\det(\mathbf{AB}) = \sum_{|J|=m} \det(\mathbf{A}_{:,J}) \det(\mathbf{B}_{J,:}).$$

Origin: H Thm. 13.8.2; S Prop. 3.4

Solution. Write the columns of $\mathbf{A}$ as $\mathbf{a}_1, \dots, \mathbf{a}_n$. Column k of $\mathbf{AB}$ is $\sum_{j=1}^n b_{jk} \mathbf{a}_j$. Multilinearity gives

$$\det(\mathbf{AB}) = \sum_{j_1, \dots, j_m} b_{j_1 1} \cdots b_{j_m m} \det(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_m}).$$

Terms with repeated indices vanish. Fix $J = \{i_1 < \dots < i_m\}$. The terms using exactly these indices are indexed by $\sigma \in S_m$ and sum to

$$\begin{aligned} \det(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_m}) & \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) \prod_{k=1}^m b_{i_{\sigma(k)}, k} \\ & = \det(\mathbf{A}_{:,J}) \det(\mathbf{B}_{J,:}). \end{aligned}$$

Sum over all J . 

Exercise 3.14 (Determinant under recoding). For square $\mathbf{S}, \mathbf{A}$ of the same order, prove

$$\det(\mathbf{SAS}') = \det(\mathbf{S})^2 \det(\mathbf{A}).$$

Deduce that orthogonal recoding leaves this determinant fixed, while rescaling every coordinate by c multiplies it by c^{2n} .

Origin: Course formulation from H Lemma 13.2.1 and Thm. 13.3.4

Solution. Multiplicativity and transpose invariance give

$$\det(\mathbf{SAS}') = \det(\mathbf{S}) \det(\mathbf{A}) \det(\mathbf{S}') = \det(\mathbf{S})^2 \det(\mathbf{A}).$$

For orthogonal $\mathbf{S}$, $\det(\mathbf{S})^2 = 1$. For $\mathbf{S} = c\mathbf{I}_n$, $\det(\mathbf{S})^2 = (c^n)^2 = c^{2n}$. 